



Search for life on Mars

The NASA 2020 Rover will search for signs of existing or pre-existing extraterrestrial life.
<https://digit.in/dec18-71-1>



RRM3 revitalizes spacecrafts

NASA's RRM3 or Robotic Refueling Mission 3 will extend spacecraft life and duration in space.
<https://digit.in/dec18-71-2>

peeking out from behind the shield, at an angle, are the solar panels. Only a small portion of the solar panels are exposed to the Sun, and they are cooled internally by water and a radiator. Sensors along the panels ensure that the panels are not over-exposed to the Sun, and re-orient the spacecraft automatically if the panels are likely to be damaged.

While scientists do know that the corona is much hotter than the surface of the Sun, they do not know why this is the case. One of the fundamental mysteries that the Parker Solar Probe hopes to solve, is to understand why the corona gets so hot. The probe will also study the phenomenon of solar wind - a stream of high energy particles that move all the way to the orbit of Neptune. These particles, cause the aurorae on Earth when they react with the atmosphere. Observations by previous spacecraft have indicated two types of solar wind - there is a slow stream and a fast stream. Mariner 2 identified the source of the fast stream as relatively cooler regions on the surface of the Sun, known as coronal holes (not to be confused with sun spots). The source of the particles in the slow stream, is not yet known. This is another mystery of the corona that the Parker Solar Probe hopes to unlock.

The problem with studying Solar Wind so far, has been that most of the observations have been taken from instruments on the Earth or close to the Earth. The Parker Solar Probe repeatedly diving into the corona will allow scientists to get up close and personal with the Sun, and better study it.

THE EXPERIMENTS ON BOARD

There are seven Venus flybys scheduled, each of which will shrink the orbit of the Spacecraft. If everything goes as planned, at its final close approach on 12 December, 2025, the probe will be just 6.9 million kilometres from the the Sun, and will be flying at 192 kilometres per second.

That is fast enough to go from New Delhi to Agra in one second. At that point, the Parker Solar Probe will set its final record for being the fastest spacecraft, as well as the the closest a man made object has got to the Sun. The probe carries four instrument suites on board, to study the Solar Wind and the Solar corona in unprecedented detail.

The Solar Wind Electrons Alphas and Protons (SWEAP) instrument is the part of the probe that will actually "touch" the Sun. A special instrument, called the Solar Probe Cup will look at the Sun directly, and capture high energy plasma particles to study. The cup is designed to capture charged particles in the vacuum of space. This instrument looks around the protective carbon composite shield, that protects most of the instruments on board from the furious energies pouring out of the Sun.

The Integrated Science Investigation of the Sun (IS²S) instrument, pronounced "eesis" has the symbol for the Sun incorporated into its acronym. The instrument has a series of solid state detectors, to record the energy levels and the movements of charged particles. These particles can be electrons, protons or ions. Researchers hope to use the instrument to study the particles that are accelerated in the Solar corona, as well as the particles that are accelerated in interplanetary space. By the time these high energy particles reach the Earth, they cannot really be differentiated, and their origins are a mystery. The instrument will study the life spans of the particles, and track their movement from the Sun and outwards into the Solar System.

Wide-Field Imager for Solar Probe Plus (WISPR) is essentially a really fancy camera. It is the only imaging instrument on board the spacecraft. The data gathered by WISPR will allow researchers to create 3D reconstructions of the shape of the Solar corona, which will help understand the context of the observations of the other instruments. WISPR will be

able to study coronal mass ejections, or events more commonly known as Solar flares.

Finally, there is the FIELDS Experiment. The instrument has antennas extending beyond the protective shield. Three magnetometers and five voltage sensors, combined with an electronics package make up the instrument. FIELDS will take measurements of the electric and magnetic fields, measure the electron density, and capture information about the temperature. The instrument is designed to measure the chaotic movements within the inner heliosphere, and track realigning magnetic fields on the Sun. FIELDS may even discover previously unknown phenomenon within the heliosphere, as it is an unexplored region of space, which we do not know much about.

As humans deploy an increasing number of satellites and astronomical instruments in space, it is necessary to understand and predict this solar wind. The solar wind is a threat to our technologies, and we stand to benefit by understanding it better. The Parker Solar Probe will be able to observe the particles accelerate from subsonic to supersonic speeds. The experiments will be conducted in one of the most extreme and hazardous regions in the Solar System. In short, the spacecraft will study the corona, and give scientists a better understanding of the source of Solar wind.

The Parker Solar Probe is the first NASA spacecraft to be named after a living person. Initially, the spacecraft was called the Solar Probe Plus, but was renamed to honour Eugene Parker. Parker developed a theory in 1958, that predicted the existence of Solar Wind. According to the theory, there is so much heat in the corona, that the high energy particles escape from the immense gravity of the Sun, and flow outwards into the Solar System. This solar wind affects all the planets in the Solar System, which is why scientists want to better understand it. The Parker Solar Probe is humanity's first mission to touch a star. 